

Evaluation Hardening Guide

Scientific Data Analysis v2.2

A compact, assessable backend compatible with Scientific Workbench

Maintenance: a clean active bundle, historical routing stub, regression wrappers, and
post-sync plugin-eval

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Translation of the preserved private v2.2 guide at commit 83dd0a5. Original TeX SHA-256:
9f9dd5ab069c07b65ba905586de2937caf4f13d024f1c80eec58afe3fde62a95. Private source:
skills/scientific-data-maintainer/.cache/active_references/guides/guia_scientific_
data_analysis_v2_2.tex.

Recorded results and closure steps are historical, not current verification. This release guide does not
synchronize or modify Scientific Workbench; the app remains outside its scope. Historical commands
are retained for traceability: use fresh output directories when reproducing them, without
overwriting earlier evidence. The original remains unchanged privately.

Abstract

`scientific-data-analysis` v2.2 hardens evaluation on top of the v2.1 compact router and integrated v2.0 baseline. Its purpose remains reproducible, non-destructive, traceable scientific and technical workflows. Packaging and evaluator visibility change: temporary files, caches, and substantial historical routing are excluded from the active bundle; representative regressions become more visible to generic tools; and the guide advances to v2.2.

The skill remains the deterministic local backend, with Scientific Workbench as the compatible app. v2.2 neither edits nor synchronizes the app.

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1. Executive overview

v2.2 adds no capability. It makes the installed skill easier to assess, maintain, and load. The recorded release work:

- Keeps `SKILL.md` as a compact router.
- Preserves the registry, JSON contracts, run bundles, artifact types, and public scripts.
- Removes `tmp/`, `__pycache__`, and historical outputs from the active bundle.
- Replaces long v2.1 routing with a compatible stub.
- Replaces internal `README.md` documents counted by `plugin-eval` as scattered documentation.
- Archives historical notes without live references in `fixtures/archived_references/`.
- Extends `unittest` wrappers for routes and app-facing failures.
- Prepares a realistic, non-destructive `plugin-eval` benchmark.
- Reruns `plugin-eval analyze` before and after synchronization.

This is v2.2, not v2.3 or v3.0. It changes neither scientific behavior nor the app, and makes no claim of unverified capabilities. The historical cleanup described here is not an instruction to delete preserved evidence today.

2. From v2.1 to v2.2

Area	v2.2 change
Active bundle	Temporary files, caches, and audit residue stay outside the installed skill.
Historical routing	<code>references/v2-1-expanded-skill-routing.md</code> becomes a lightweight stub.
Internal documents	Non-primary <code>README.md</code> files become <code>README.txt</code> .
Historical archive	Notes not linked from live routes move to <code>fixtures/archived_references/</code> .
Visible regressions	<code>tests/test_audit_regression_wrappers.py</code> covers audits, public surface, general routes, FITS, and clean failures.
Plugin-eval	Records a baseline, measures deferred cost, and reassesses after synchronization.
Guide	Adds <code>guia_scientific_data_analysis_v2_2.tex</code> and <code>.pdf</code> .

3. What remains unchanged

- No capability renaming or changes to `public_surface_registry.yaml`.
- No changes to frozen artifact types or scientific script contracts.
- Optional backends do not become core dependencies.
- Scientific Workbench is neither edited nor synchronized.

4. Evaluation rules

Evaluate the installed skill with `plugin-eval`. These historical command templates use a placeholder for the evaluator location; resolve that location before use. The original account-specific home path is replaced with `$HOME`.

```
node .../plugin-eval.js analyze "$HOME/.codex/skills/scientific-data-analysis" --format
  markdown
node .../plugin-eval.js explain-budget "$HOME/.codex/skills/scientific-data-analysis" --
  format markdown
```

Interpret the measurements accurately:

- `trigger_cost_tokens`: cost of activating the skill.
- `invoke_cost_tokens`: cost of reading initial instructions.
- `deferred_cost_tokens`: bundled material that should not always be loaded.
- Python warnings are real technical debt, but do not all justify a large refactor during a packaging release.

5. Historical v2.2 closure criteria

Gate	Criterion
<code>py_compile</code>	Changed scripts and wrappers compile.
<code>sync_public_surface_docs.py -check</code>	Registry and snapshot remain consistent.
<code>skill_surface_audit.py</code>	Public surface remains valid.
<code>unittest wrappers</code>	Representative regressions pass.
<code>portable_smoke_test.py -profile core</code>	Core smoke passes or blocks in a controlled, justified way.
<code>plugin-eval analyze</code>	Score improves or remaining warnings are documented.
<code>benchmark.json</code>	Concrete scenarios are ready; execution may be manual if cost/time is unreasonable.
<code>post-sync</code>	Installed validation matches editable validation.

6. Archive policy

The original guide records preservation outside the skill root at the following historical location, shown with a portable alias:

```
$PRIVATE_WORKSPACE/skill_archives/scientific-data-analysis-v2_2-evaluation-hardening
-20260617_100731
```

`$PRIVATE_WORKSPACE` identifies the original private editable workspace. This archive supports recovery and traceability; it is not a public contract. Do not restore the entire archive into the installed skill.

7. Normal usage after v2.2

End-user usage remains unchanged:

- General tables: `profile_table.py`, `cross_domain_data_workbench.py`.
- Document intake: `document_intake_workbench.py`.
- Astronomy/FITS: `inspect_fits.py`, `fits_rgb_batch.py`, `photometry_noise_budget.py`.

- Notebooks: `notebook_workbench.py`.
- App-readiness: JSON envelopes, run bundles, and artifact types remain valid.

8. Accepted debt

The original release accepts warnings that do not justify a large v2.2 refactor:

- Some substantial scientific scripts still have high complexity.
- Some long lines originate in contracts, tables, or app-facing messages.
- Deep deduplication belongs to an architecture phase rather than an evaluation release.

Do not break correct scientific behavior merely to improve an automated rating.

9. Recommended commands

```
python3 -m py_compile scripts/*.py scripts/_internal/*.py tests/
test_audit_regression_wrappers.py
python3 scripts/sync_public_surface_docs.py --check
python3 scripts/skill_surface_audit.py --skip-hygiene
python3 -m unittest tests.test_audit_regression_wrappers
python3 scripts/portable_smoke_test.py --profile core
```

10. Conclusion

v2.2 improves evaluation and maintenance while preserving the central purpose: reproducible local-first analysis, original preservation, traceable outputs, and compatibility with Scientific Workbench.